

Places to Visit Guide

Project Proposal · WDD330 · Author: Erick Scala

1. General Description

Problem: When planning a trip, it can be difficult to discover what is near a destination, filter results by place type, and view locations on a map all in one place.

Solution: "Places to Visit Guide" is a web application where users enter a city or address (or use their current location) to get a list and an interactive map of nearby tourist attractions.

Motivation: I am passionate about traveling and discovering off-the-beaten-path locations, and I have always found it challenging to find—in a single place—what to visit at a destination and where those spots are located. Additionally, this project is an ideal opportunity to strengthen a skill I have not yet mastered: consuming REST APIs.

2. Target Audience

Travelers and tourists wishing to discover places to visit in a new destination.

Families and groups planning weekend outings in their local area.

Students and budget-conscious travelers looking for free options that do not require creating an account.

Mobile users checking the guide while exploring the city.

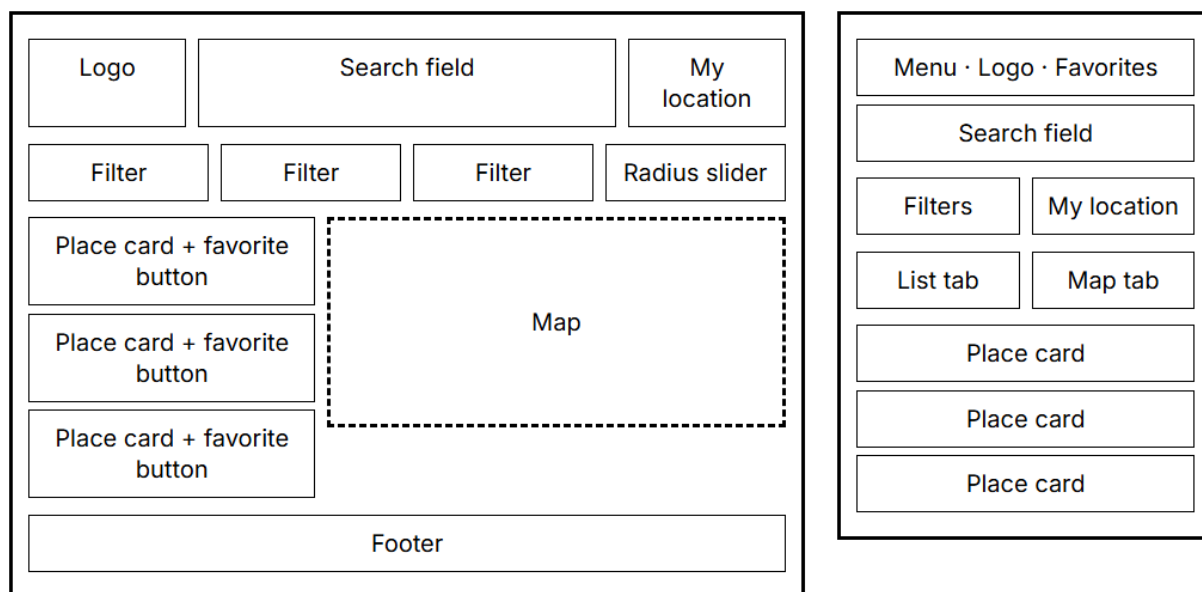
3. Major Functions

N°	Feature	Description
1	Location search	Search field for a city, country, or address. Uses Nominatim to convert the text into coordinates (the first result is used). The search runs when the user presses Enter or the search button.

2	Interactive map	Map (Leaflet + OpenStreetMap tiles) centered on the destination, with a marker for each point of interest and the required “© OpenStreetMap contributors” credit.
3	Attractions list	Cards showing name, category, distance to the center, and popularity. Clicking a card highlights its corresponding marker.
4	Category filter	Buttons/chips to filter by: museums, architecture, nature, religion, history, entertainment.
5	Place details	Modal view or page featuring an image, description, address, Wikipedia link, and a "get directions" button (OpenStreetMap link).
6	Favorites	Save/remove places using a heart icon. Stored in localStorage with a dedicated view.
7	Use my location	Button using the browser's Geolocation API to search for attractions near the user.
8	Error and state handling	Loading indicators; messages for no results, connection issues, or API limit reached.

4. Wireframes

View 1 – Home and results (desktop and mobile)



View 2 – Place details

Back button · Place name · Favorite button	
Image	Category · Rating
	Address
	Wikipedia link · Directions link
Description	
Mini map	

Back · Name · Favorite
Image
Category · Rating
Description
Address
Mini map
Directions link

View 3 – Favorites and history

Logo · Search · Home link		
Favorites title		
Place card	Place card	Place card
Place card	Place card	Place card
Recent searches (list of links)		

Menu · Favorites title
Place card
Place card
Place card
Recent searches

5. External Data

Source	Purpose
<u>OpenStreetMap</u> Nominatim	Geocoding: converts a typed city or address into latitude/longitude and returns the full place name.
OpenTripMap	Retrieves tourist attractions within a fixed 5 km radius and the details of each one.
Leaflet + OSM tiles	Map rendering and markers (a mapping library, not a data API).

Data to store (localStorage):

- favorites: array of saved places (xid, name, category, coordinates, image).

6. Module List

Module	Responsibility
api.mjs	Nominatim and OpenTripMap requests: city search, list by category, details by xid, and error handling.
map.mjs	Initializes Leaflet, adds/removes markers, syncs the map with the list.
ui.mjs	Renders cards, detail modal, filter chips, loaders, and messages.
storage.mjs	Reads/writes favorites in localStorage.
main.mjs	Entry point: wires up events, handles geolocation, and switches between views.

7. Graphic Identity

Color scheme

Teal #0F6B6B (primary)
Orange #E8743B (accent/CTA)
White #FFFFFF (background)
Dark gray #1F2A2E (text)

Typography

- Inter 400/500 (single font for headings and body)
- Fallback: system-ui, sans-serif

Element styling

- Cards with 10 px rounded corners
- Rounded buttons; orange for main calls to action
- Filter chips turn teal when active
- Simple line icons; heart icon for favorites

App icon

A rounded teal square containing a white location pin. Inside the pin is an orange circle with a compass needle, symbolizing the discovery of new places.

8. Timeline (Weeks 5–7)

Week	Focus	Deliverables
5	Foundation and data	Project structure and repository; basic responsive HTML/CSS; OpenTripMap API key; city search with Nominatim; list of attractions from OpenTripMap.
6	Map and interaction	Leaflet map with markers synced to the list; category filter; place detail view; loading and error messages.
7	Persistence and polish	Favorites with localStorage and favorites view; “Use my location” button; final styling and basic accessibility;

		testing on mobile and desktop; deployment on GitHub Pages; final presentation.
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9. Project Planning (Trello)

Board link: <https://trello.com/b/bGgvGUAd/places-visit-guide>

To Do

- Get OpenTripMap API key
- Test endpoints
- Create repository and folder structure
- Define color palette and typography in CSS
- Leaflet map + markers
- Category filter
- Place detail view
- Favorites with localStorage
- Geolocation
- Mobile testing
- Deploy to GitHub Pages

Bugs / Improvements

- (To be filled in during development)

10. Challenges

- **API limits and keys:** Nominatim is limited to 1 request per second and OpenTripMap has a free quota. I will search only on Enter or button click and show clear messages if a limit is reached.
- **Incomplete data:** many places lack images, descriptions, or ratings, so I need default values and layouts that do not break.

- **Syncing map and list:** keeping markers, cards, and filters consistent when the search or filter changes.
- **Asynchronous flow and errors:** chaining two APIs (geocode first, then fetch places) with async/await and handling network failures.